Physical Chemistry (I)

Lecture 23

Electrolytes and Electrodes

In the Last Class

- Thermodynamics of electrochemical cells
  - Generalization of thermodynamics
  - Thermodynamics of cell potential

- Physical chemistry of electrolytes
  - Gibbs energy for solvation: Born equation
  - Potential distributions: Debye-Hückel equation

Generalization

For a non-expansion work $\delta w = f\mathrm{d}X$,

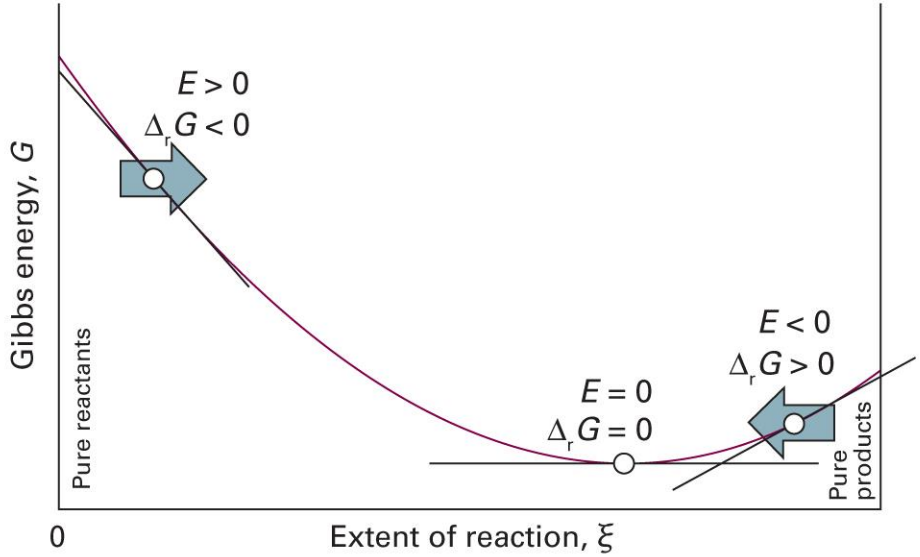
- State: $(p, T, n_\mathrm{A}, n_\mathrm{B}, \cdots) \rightarrow (p, T, X, n_\mathrm{A}, n_\mathrm{B}, \cdots)$

- Thermodynamic potentials
  - $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + f\mathrm{d}X + \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} + \cdots$
  - $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p + f\mathrm{d}X + \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} + \cdots$
  - $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + f\mathrm{d}X + \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} + \cdots$
  - $\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + f\mathrm{d}X + \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} + \cdots$

Last class

Cell Potential and Reaction

$$\Delta_r G = -\nu F E_{cell}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 6C.4)

Nernst Equation

$$E_{\text{cell}} = E^{\ominus}_{\text{cell}} - \frac{RT}{\nu F}\ln Q$$

$$E^{\ominus}_{\text{cell}} = \frac{RT}{\nu F}\ln K$$

Born Equation

$$\Delta_{\mathrm{solv}}G^{\ominus} = -\frac{z_i^2 e^2}{8\pi\varepsilon_0 r_i}\left(1 - \frac{1}{\varepsilon_{\mathrm{r}}}\right)$$

where the relative permittivity $\varepsilon_{\mathrm{r}} = \varepsilon/\varepsilon_0$.

- $\Delta_{\mathrm{solv}}G^{\ominus} < 0$: solvation of ions is favored
- Water has ε_{r} = 78.54 at 25°C, and

$$\Delta_{\mathrm{solv}}G^{\ominus} = -\frac{z_i^2}{r_i/\mathrm{pm}} \times (6.86 \times 10^4\ \mathrm{kJ\ mol^{-1}})$$

Last class

Debye-Hückel Equation

- Poisson-Boltzmann equation

$$\nabla^2 \phi = \frac{zFn_\infty}{\varepsilon V}\left(e^{+ze\phi/k_\mathrm{B}T} - e^{-ze\phi/k_\mathrm{B}T}\right)$$

- Debye-Hückel equation

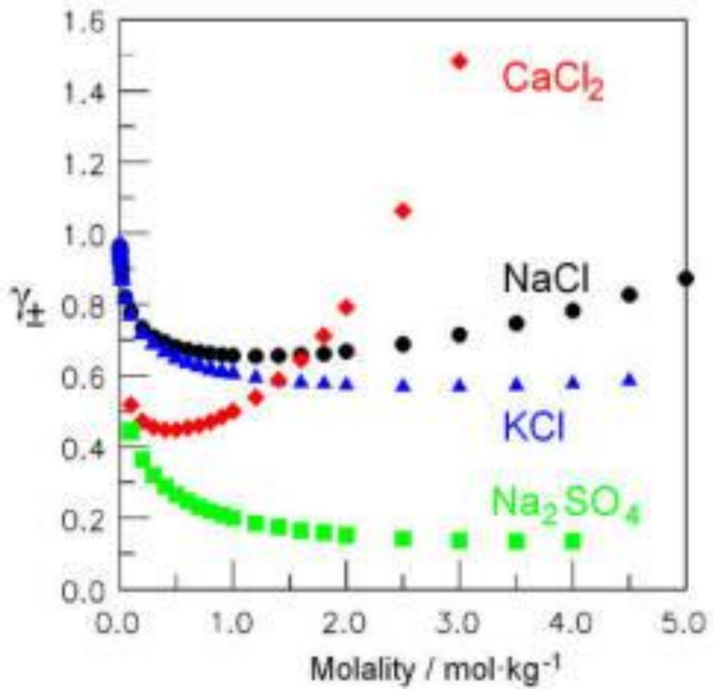
$$\nabla^2 \phi = \phi / r_\mathrm{D}^2$$

where the Debye radius r_D is

$$r_\mathrm{D} = \sqrt{\frac{\varepsilon RT}{2F^2 I b^\ominus (m_\mathrm{A}/V)}}$$

Activities of Ions

Quickly deviate from ideal behavior



(Image: https://chem.libretexts.org/@go/page/44345)

Molar Gibbs Energy of Ions

For a real solution consisting of M^+ and X^-,

$$G_\mathrm{m} = \mu_+ + \mu_-$$

However,

$$\mu_J = \mu_J^\ominus + RT\ln a_J = \mu_J^\ominus + RT\ln(\gamma_J x_J) = \mu_J^\mathrm{ideal} + RT\ln\gamma_J$$

and

$$G_\mathrm{m} = \mu_+ + \mu_- = \mu_+^\mathrm{ideal} + \mu_-^\mathrm{ideal} + RT\ln\gamma_+ + RT\ln\gamma_-$$

Hence,

$$G_\mathrm{m} = G_\mathrm{m}^\mathrm{ideal} + RT\ln(\gamma_+\gamma_-)$$

Mean Activity Coefficients

$$G_\mathrm{m} = G_\mathrm{m}^\mathrm{ideal} + RT\ln(\gamma_+\gamma_-)$$

Experimentally impossible to separate γ_+ and γ_-

- Mean activity coefficient: geometric mean of γ_+ and γ_-

$$\gamma_\pm = (\gamma_+\gamma_-)^{1/2}$$

Hence,

$$\mu_+ = \mu_+^\mathrm{ideal} + RT\ln\gamma_\pm$$

$$\mu_- = \mu_-^\mathrm{ideal} + RT\ln\gamma_\pm$$

Generalization

For a salt M_pX_q (= $pM^{|z_+|+} + qX^{|z_-|-}$),

$$G_m = p\mu_+ + q\mu_- = G_m^{ideal} + pRT\ln\gamma_+ + qRT\ln\gamma_-$$

The mean activity coefficient is defined as

$$\gamma_\pm = \left(\gamma_+^p\gamma_-^q\right)^{1/(p+q)}$$

so that

$$\mu_J = \mu_J^{ideal} + RT\ln\gamma_\pm$$

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Activity and Work

$$\gamma_\pm = \left(\gamma_+^p \gamma_-^q\right)^{1/(p+q)}$$

Since $G_\mathrm{m} = G_\mathrm{m}^\mathrm{ideal} + pRT\ln\gamma_+ + qRT\ln\gamma_-$,

$$G_\mathrm{m} = G_\mathrm{m}^\mathrm{ideal} + RT\ln\left(\gamma_+^p \gamma_-^q\right) = G_\mathrm{m}^\mathrm{ideal} + (p+q)RT\ln\gamma_\pm$$

Hence,

work of charging an ion

$$\ln\gamma_\pm = \frac{G_\mathrm{m} - G_\mathrm{m}^\mathrm{ideal}}{(p+q)RT} = \frac{w_\mathrm{e}}{(p+q)RT}$$

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Work of Charging an Ion

- The work of charging an ion:

$$\delta w_{\mathrm{e}} = \phi_{\mathrm{others}} \mathrm{d}Q$$

- The potential due to other ions:

$$\phi_{\mathrm{others}} = \phi_{\mathrm{DH}} - \phi_{\mathrm{self}} = \frac{Q}{4\pi\varepsilon r} e^{-r/r_{\mathrm{D}}} - \frac{Q}{4\pi\varepsilon r}$$

As we are interested in the vicinity of the ion ($r \to 0$),

$$e^{-r/r_{\mathrm{D}}} \approx 1 - r/r_{\mathrm{D}}$$

$$\delta w_{\mathrm{e}} = -\frac{Q}{4\pi\varepsilon r_{\mathrm{D}}} \mathrm{d}Q$$

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Work of Charging an Ion

$$\delta w_{\mathrm{e}} = -\frac{Q}{4\pi\varepsilon r_{\mathrm{D}}}\mathrm{d}Q$$

For an ion of charge $z_i e$,

$$w_{\mathrm{e}} = -\frac{1}{4\pi\varepsilon r_{\mathrm{D}}}\int_0^{z_i e} Q\,\mathrm{d}Q = -\frac{z_i^2 e^2}{8\pi\varepsilon r_{\mathrm{D}}}$$

The molar work is

$$w_{\mathrm{e,m}} = -\frac{N_{\mathrm{A}} z_i^2 e^2}{8\pi\varepsilon r_{\mathrm{D}}} = -\frac{z_i^2 F^2}{8\pi N_{\mathrm{A}}\varepsilon r_{\mathrm{D}}}$$

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Total Work

$$w_{\mathrm{e,m}} = -\frac{z_i^2 F^2}{8\pi N_{\mathrm{A}} \varepsilon r_{\mathrm{D}}}$$

The total work of charging p moles of M and q moles of X:

$$w_{\mathrm{e,total}} = p w_{\mathrm{e,m}}(\mathrm{M}) + q w_{\mathrm{e,m}}(\mathrm{X}) = -\frac{(p z_+^2 + q z_-^2) F^2}{8\pi N_{\mathrm{A}} \varepsilon r_{\mathrm{D}}}$$

Since $p z_+ + q z_- = 0$ (charge neutrality),

$$p z_+^2 + q z_-^2 = z_+(-q z_-) + z_-(-p z_+) = -(p+q) z_+ z_-$$

$$w_{\mathrm{e,total}} = \frac{(p+q) z_+ z_- F^2}{8\pi N_{\mathrm{A}} \varepsilon r_{\mathrm{D}}} = -\frac{(p+q)|z_+ z_-| F^2}{8\pi N_{\mathrm{A}} \varepsilon r_{\mathrm{D}}}$$

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Activity

$$w_{\mathrm{e,total}} = -\frac{(p+q)|z_+z_-|F^2}{8\pi N_{\mathrm{A}}\varepsilon r_{\mathrm{D}}}$$

$$\ln \gamma_\pm = \frac{w_{\mathrm{e}}}{(p+q)RT} = -\frac{|z_+z_-|F^2}{8\pi N_{\mathrm{A}}RT\varepsilon r_{\mathrm{D}}}$$

Since $r_{\mathrm{D}} = \{\varepsilon RT/2F^2 I b^\ominus(m_{\mathrm{A}}/V)\}^{1/2}$,

$$\ln \gamma_\pm = -|z_+z_-|\left[\frac{F^3}{4\pi N_{\mathrm{A}}}\left\{\frac{b^\ominus(m_{\mathrm{A}}/V)}{2\varepsilon^3R^3T^3}\right\}^{1/2}\right]I^{1/2}$$

Debye-Hückel Limiting Law

$$\ln \gamma_{\pm} = -|z_+ z_-| \left[\frac{F^3}{4\pi N_A} \left\{ \frac{b^{\ominus}(m_A/V)}{2\varepsilon^3 R^3 T^3} \right\}^{1/2} \right] I^{1/2}$$

$$\log_{10} \gamma_{\pm} = -|z_+ z_-| \left[\frac{F^3}{4\pi N_A \ln 10} \left\{ \frac{b^{\ominus}(m_A/V)}{2\varepsilon^3 R^3 T^3} \right\}^{1/2} \right] I^{1/2}$$

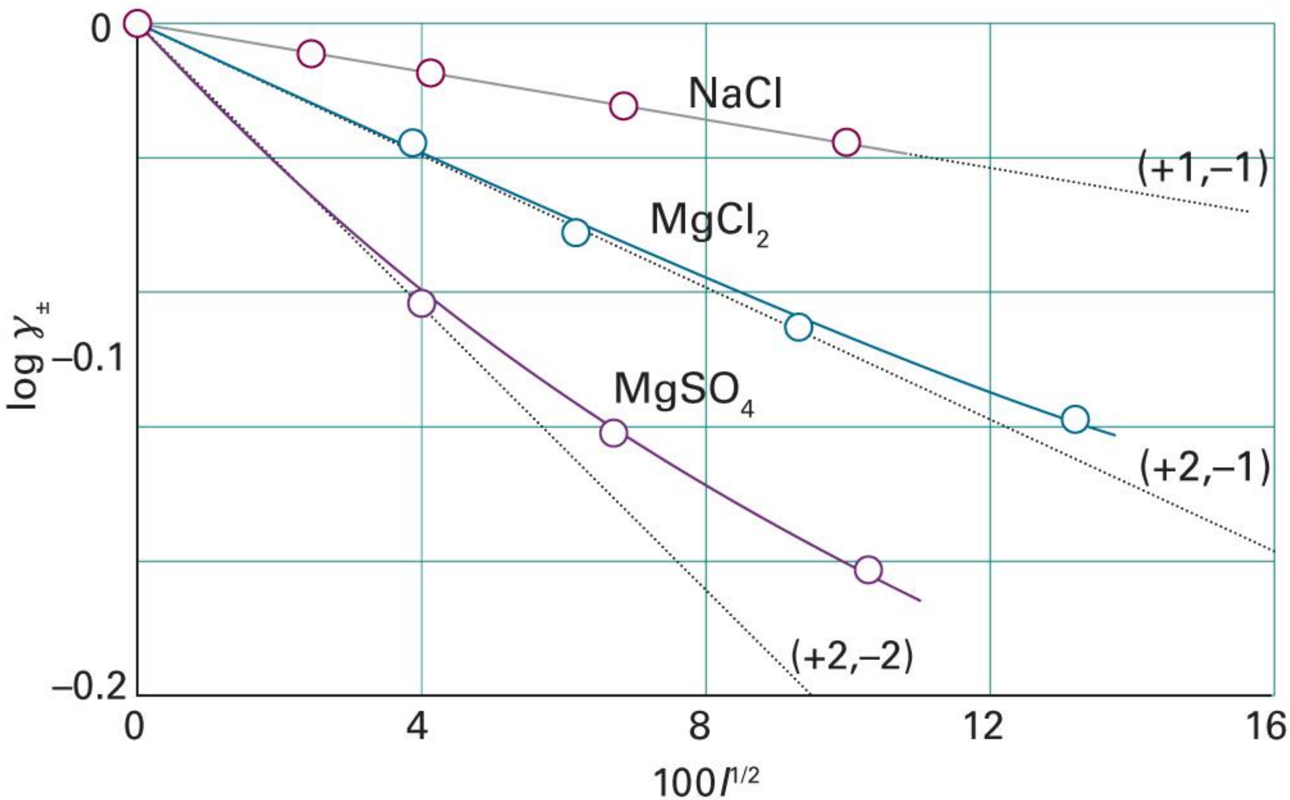
Hence,

$$\log_{10} \gamma_{\pm} = -A|z_+ z_-| I^{1/2}$$

where $A = 0.509$ for an aqueous solution at 25°C

Topic 5F

Debye-Hückel Limiting Law



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5F.5)

Extensions

- Debye-Hückel law (1923)

$$\log_{10}\gamma_{\pm} = -A|z_+z_-|I^{1/2}$$

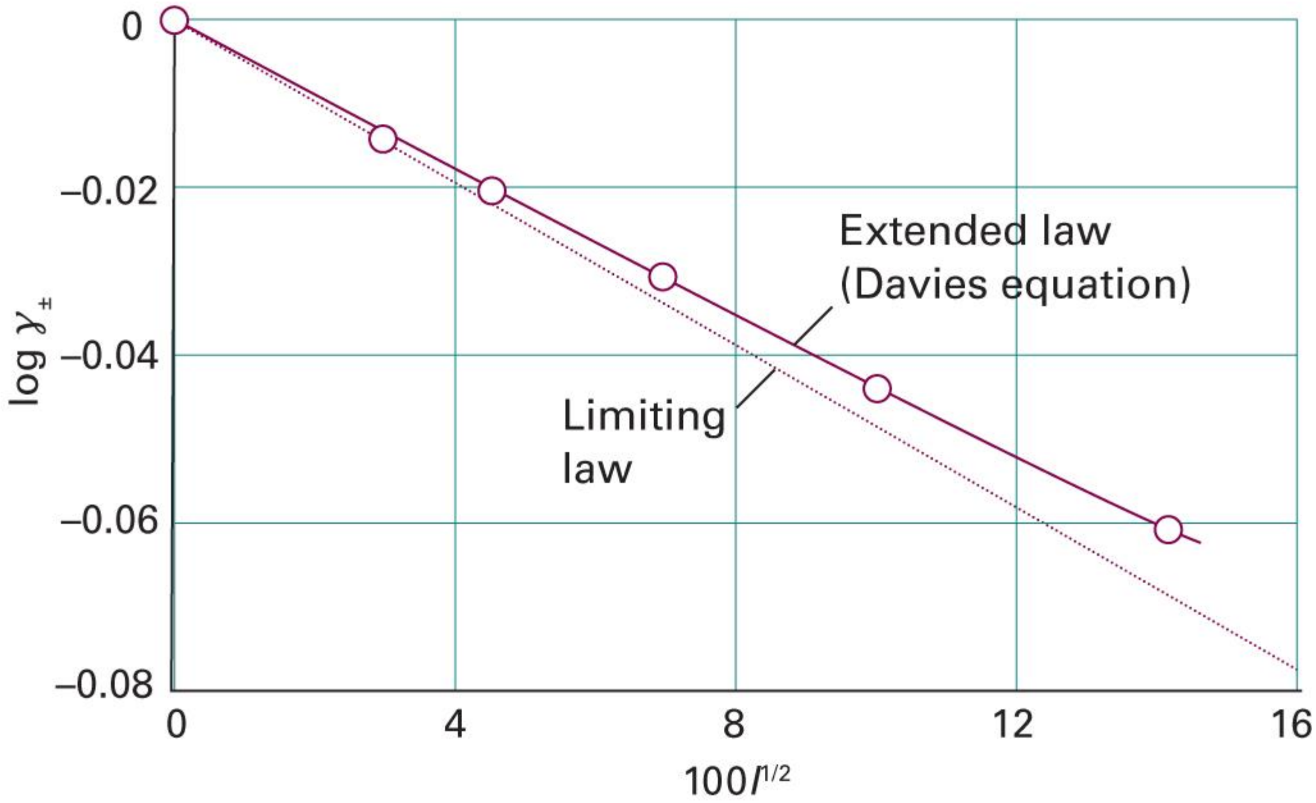
- Extended Debye-Hückel law

$$\log_{10}\gamma_{\pm} = -\frac{A|z_+z_-|I^{1/2}}{1+BI^{1/2}}$$

- Davies equation (1938)

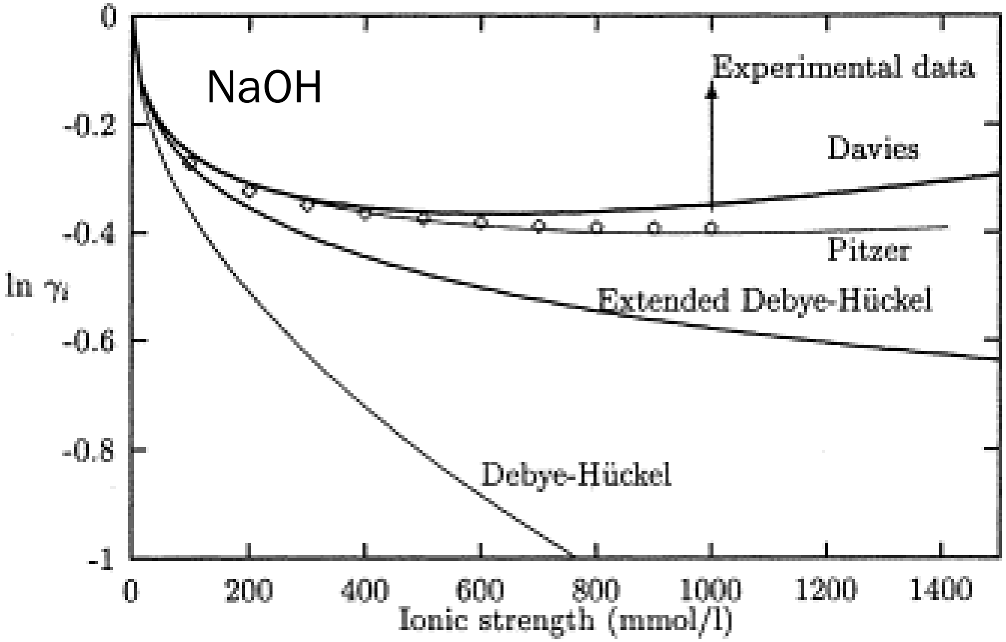
$$\log_{10}\gamma_{\pm} = -\frac{A|z_+z_-|I^{1/2}}{1+BI^{1/2}} + CI$$

Model Performances



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5F.6)

At Higher Concentrations



(Image: Samson *et al.*, ***Computational Materials Science* 15**: 285-294 (1999), Figure 1)

Experimental Determination

Consider the Harned cell:

$$\mathrm{Pt(s)|H_2(g)|HCl(}aq,b\mathrm{)|AgCl(s)|Ag(s)}$$

$$(1/2)\mathrm{H_2(g) + AgCl(s) \longrightarrow HCl(}aq\mathrm{) + Ag(s)}$$

- Cell potential

$$E^{\ominus}_{\text{cell}} = E^{\ominus}(\mathrm{AgCl/Ag,Cl^-}) - E^{\ominus}(\mathrm{SHE}) = E^{\ominus}(\mathrm{AgCl/Ag,Cl^-})$$

- Nernst equation

$$E_{\text{cell}} = E^{\ominus}(\mathrm{AgCl/Ag,Cl^-}) - \frac{RT}{F}\ln\frac{a(\mathrm{H^+})a(\mathrm{Cl^-})}{\{a(\mathrm{H_2})\}^{1/2}}$$

Activities

$$E_{\text{cell}} = E^{\ominus}(\text{AgCl/Ag,Cl}^-) - \frac{RT}{F}\ln\frac{a(\text{H}^+)a(\text{Cl}^-)}{\{a(\text{H}_2)\}^{1/2}}$$

- Let $E^{\ominus} = E^{\ominus}(\text{AgCl/Ag,Cl}^-)$
- At the standard pressure of 1 bar, $a(\text{H}_2) = 1$
- For the molality b of HCl(*aq*),

$$a(\text{H}^+) = \gamma_{\pm}b/b^{\ominus} \text{ and } a(\text{Cl}^-) = \gamma_{\pm}b/b^{\ominus}$$

$$E_{\text{cell}} = E^{\ominus} - \frac{RT}{F}\ln\left(\frac{\gamma_{\pm}b}{b^{\ominus}}\right)^2$$

Molality Dependence

$$E_{\text{cell}} = E^{\ominus} - \frac{RT}{F}\ln\left(\frac{\gamma_{\pm} b}{b^{\ominus}}\right)^2 = E^{\ominus} - \frac{2RT}{F}\left(\ln\gamma_{\pm} + \ln\frac{b}{b^{\ominus}}\right)$$

From the Debye-Hückel limiting law,

$$\ln\gamma_{\pm} = (\ln 10)\log\gamma_{\pm} = -(\ln 10)A|z_{+}z_{-}|I^{1/2}$$

$$= -(A\ln 10)(b/b^{\text{o}})^{1/2}$$

$$E_{\text{cell}} + \frac{2RT}{F}\ln\frac{b}{b^{\ominus}} = E^{\ominus} - \frac{2RT}{F}\ln\gamma_{\pm} = E^{\ominus} + \frac{2ART\ln 10}{F}\sqrt{\frac{b}{b^{\ominus}}}$$

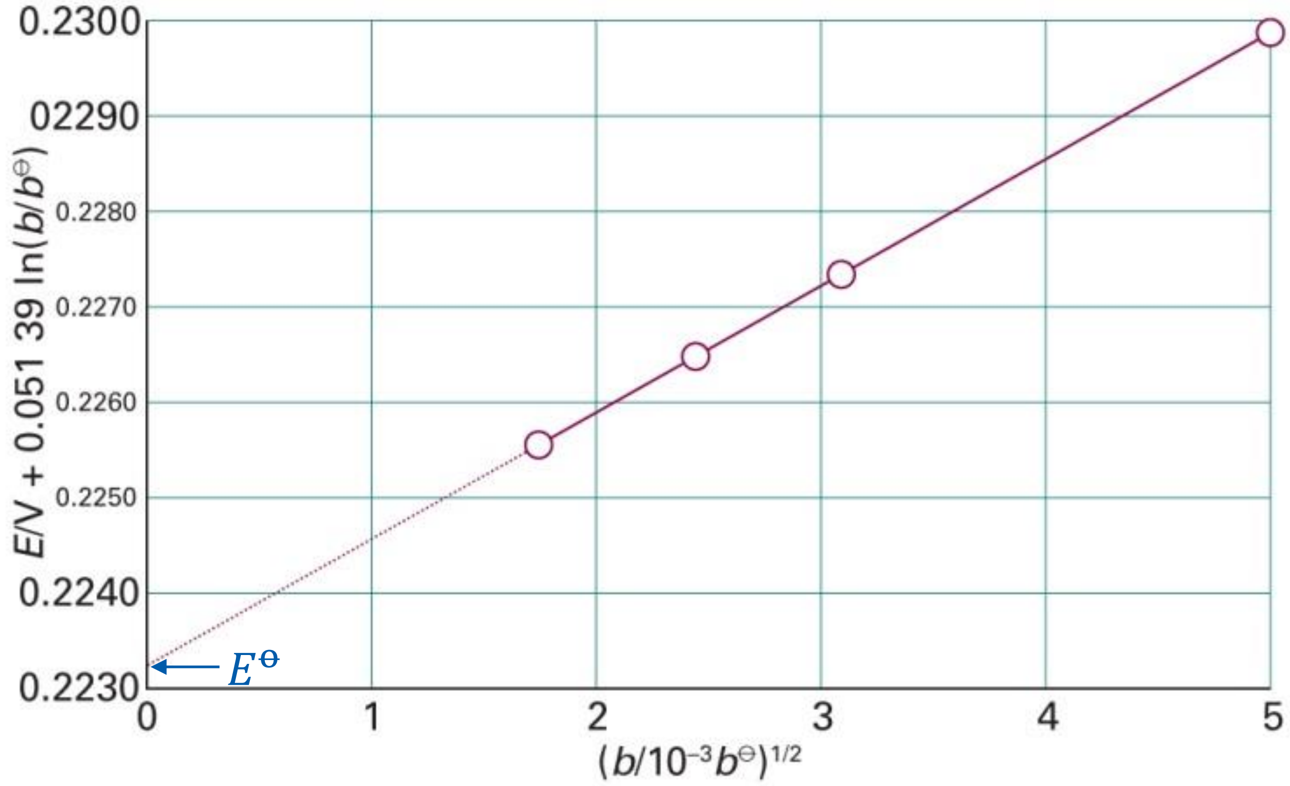
Determination of $E^{\ominus}$

$$\underbrace{E_{\text{cell}} + \frac{2RT}{F}\ln\frac{b}{b^{\ominus}}}_{y} = E^{\ominus} + \underbrace{\frac{2ART\ln 10}{F}}_{a}\underbrace{\sqrt{\frac{b}{b^{\ominus}}}}_{x}$$

From the plot of x versus y,

we can obtain $E^{\ominus}$ as the y-intercept

Example



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 6D.1)

Determination of $\gamma_{\pm}$

$$E_{\text{cell}} + \frac{2RT}{F}\ln\frac{b}{b^{\ominus}} = E^{\ominus} - \frac{2RT}{F}\ln\gamma_{\pm}$$

From the measurement of E_{cell}, we can obtain

$$\ln\gamma_{\pm} = \frac{F}{2RT}(E^{\ominus} - E_{\text{cell}}) - \ln\frac{b}{b^{\ominus}}$$

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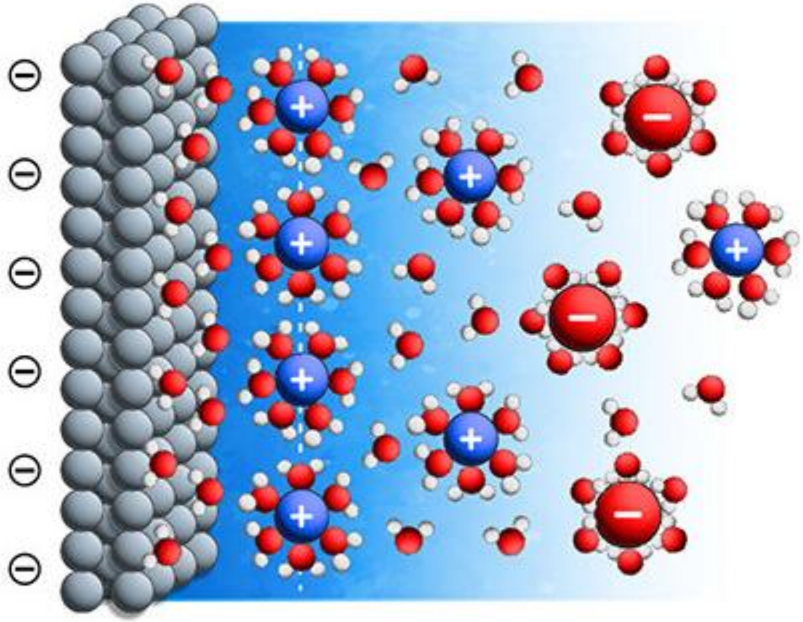
Electrode-Ion Interface



(Image: Coskun *et al*., ***Langmuir* 40**: 3283-3300 (2024), Figure 4)

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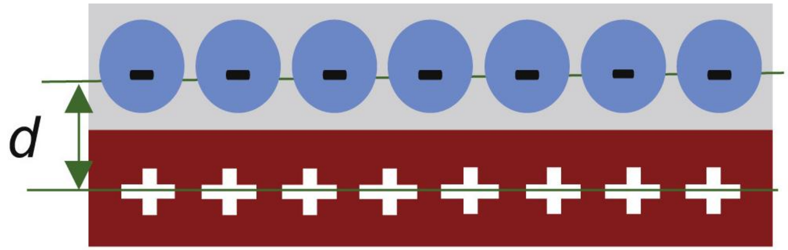
Helmholtz Model



(Image: Tao *et al*., *Green Energy & Environment* 5: 303-321 (2020), Figure 1)

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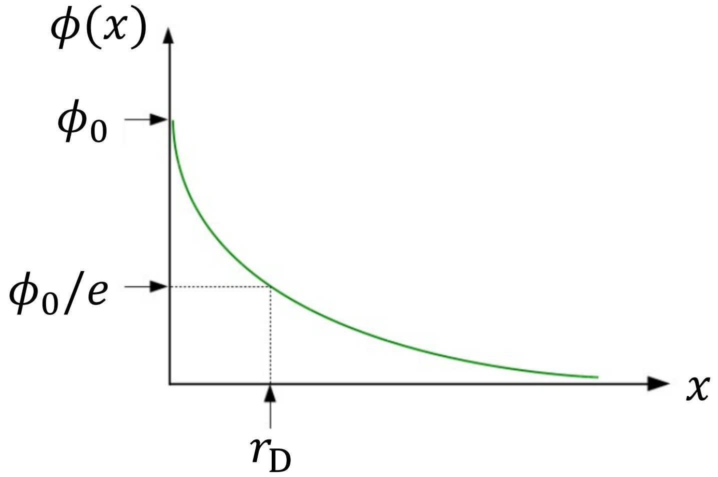
Debye-Hückel Equation

For a charged plate, solving the differential equation

$$\nabla^2 \phi = \phi / r_{\mathrm{D}}^2$$

gives a solution of

$$\phi(x) = \phi_0 e^{-x/r_{\mathrm{D}}}$$



(Image: https://docplayer.net/21695464-Charged-interfaces-formation-of-charged-surfaces-helmholz-gouy-chapman-and-stern-model-of-double-layer.html)

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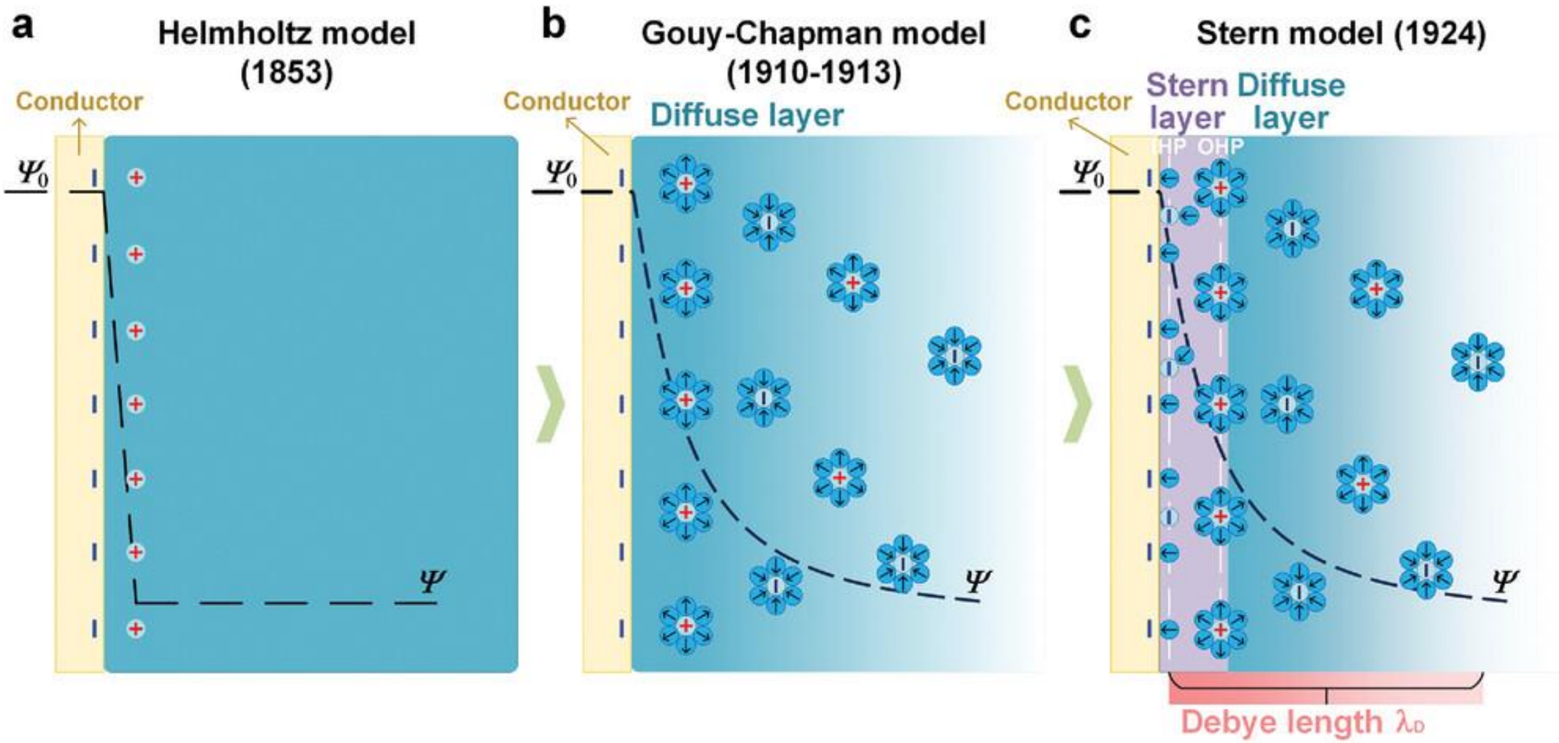
Electric Double Layer (EDL)



(Image: Li *et al.*, *Advanced Functional Materials* **34**: 2405520 (2024), Figure 2)

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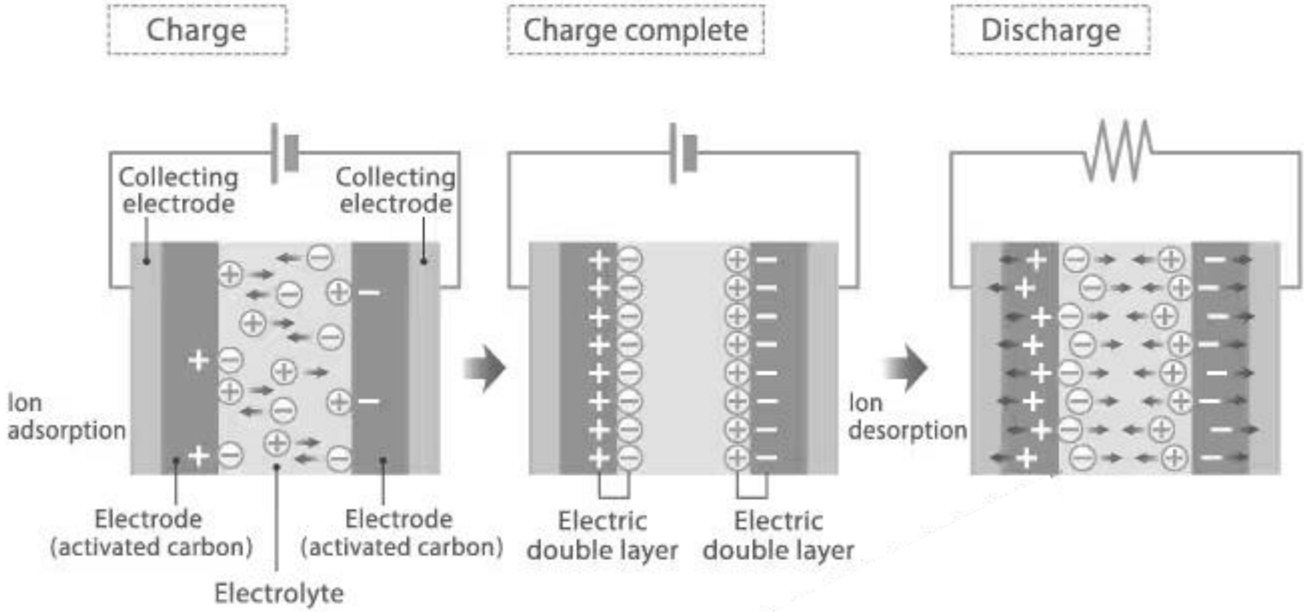
EDL Capacitor



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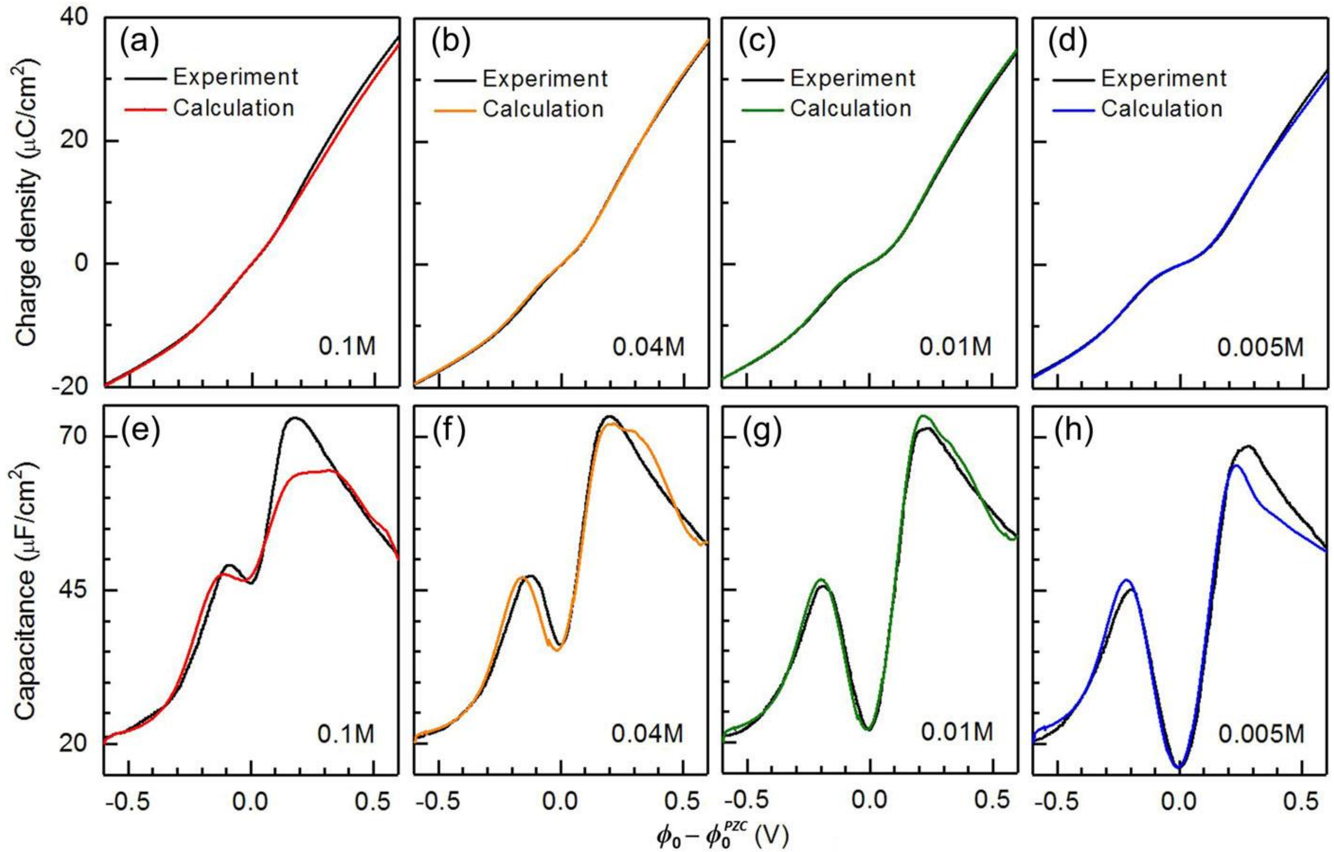
Capacitance Measurement

Ag (111) – NaF(*aq*) system



(Image: Wang *et al.*, ***Journal of Chemical Physics* 154**: 124701 (2021), Figure 5)

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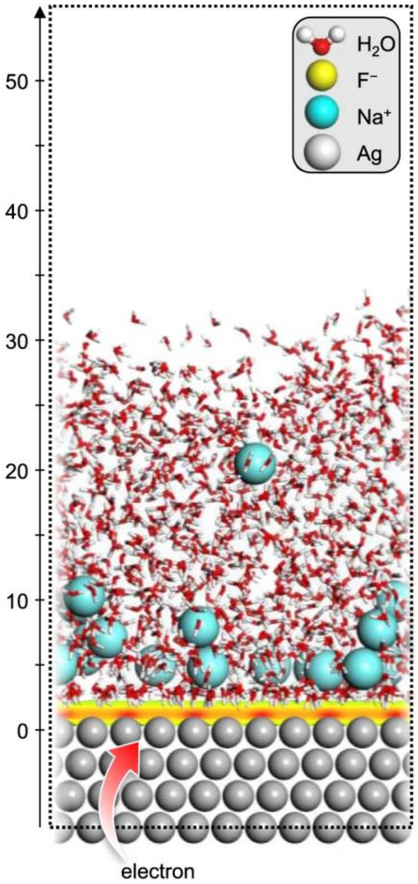
Computer Modeling



(Image: Shin *et al.*, ***Nature Communications* 13**: 174 (2022), Figure 1)

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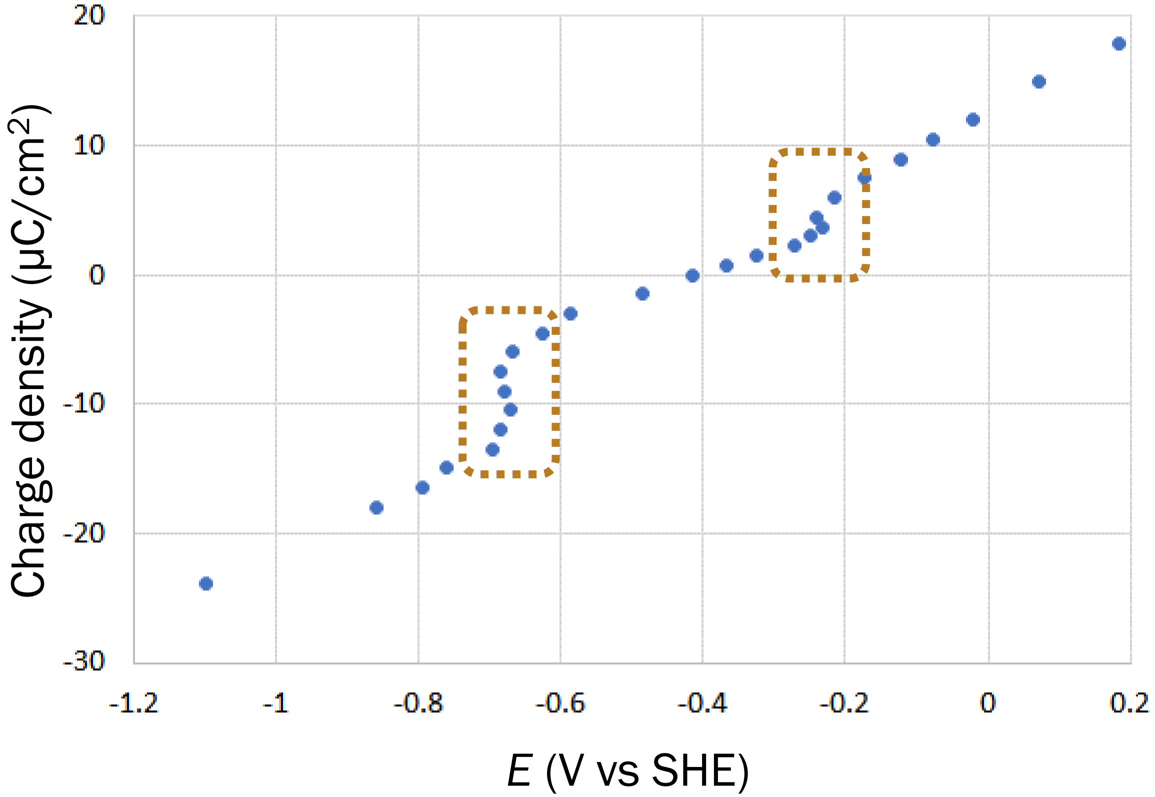
Computational Results



(Data: Shin *et al.*, *Nature Communications* **13**: 174 (2022))

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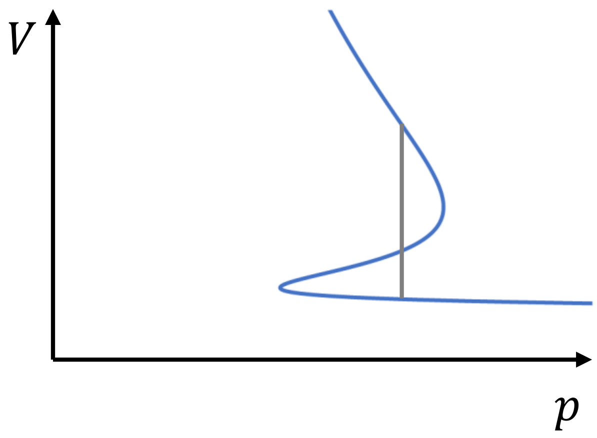
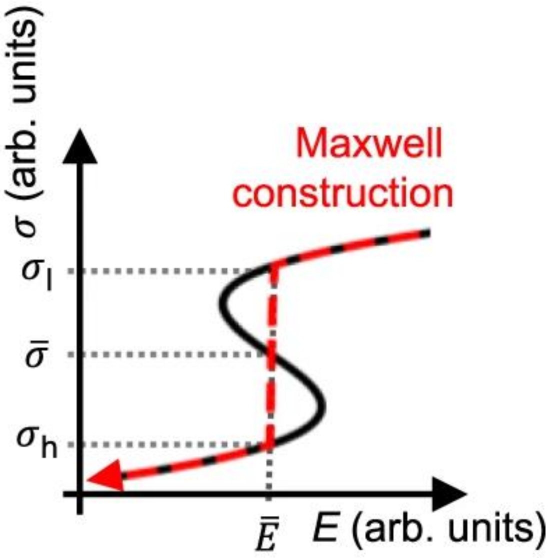
Phase Transition!





(Image: Shin *et al.*, ***Nature Communications* 13**: 174 (2022), Figure 3)

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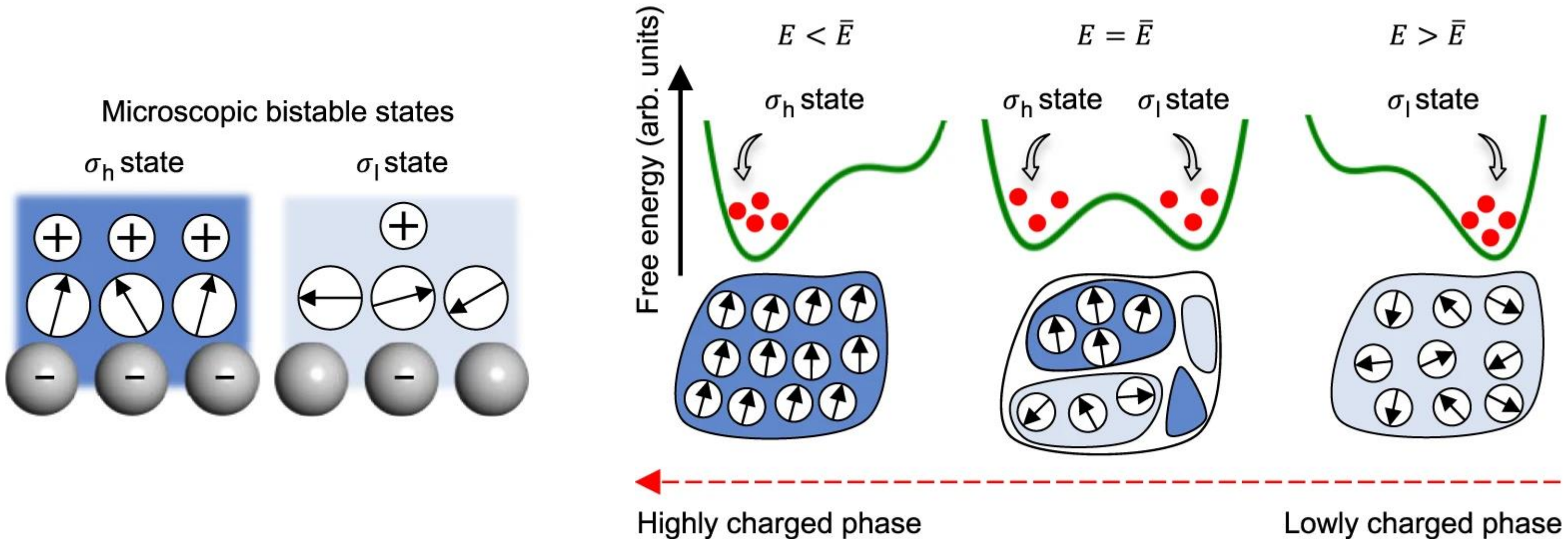
Molecular Details



(Image: Shin *et al.*, ***Nature Communications* 13**: 174 (2022), Figure 3)

Summary

Ions (electrolytes)

$$\delta w_{\mathrm{elec}} = \phi \mathrm{d}Q$$

- Poisson equation
- Boltzmann distribution

Solvation

Born equation

Electric potentials

Poisson-Boltzmann eq.

Debye-Hückel eq.

- Ion strength
- Debye radius

Activity

Debye-Hückel law

- Determination of $\gamma_{\pm}$

Ion-electrode interface

- Electric double layer

$$\mathrm{d}G_{T,p} = \delta w_{\mathrm{add,max}}$$

Electrochemical cell

- Cell potential

$$\Delta_{\mathrm{r}} G = -\nu F E_{\mathrm{cell}}$$

- Nernst equation

$$E_{\mathrm{cell}} = E_{\mathrm{cell}}^{\ominus} - \frac{RT}{\nu F} \ln Q$$

- Electrode potentials